

BRANDED PRODUCT DOCUMENTATION



AI-Powered Restaurant Financial Management SaaS Platform

Professional Project Report | Final Year Major Project | Startup Prototype Edition

Architecture
MERN + AI Layer

Business Scope
Finance + Inventory
+ Analytics

Delivery Model
Multi-tenant SaaS

1. Project Overview

This platform is a unified restaurant operations and financial intelligence system designed to replace fragmented processes with a connected SaaS workflow. It captures every transaction, expense, and inventory movement in one place and converts raw records into actionable analytics.

Problem solved: data fragmentation, manual errors, delayed decisions, and weak inventory visibility.

Core Objectives

- Track revenue and orders in real time
- Manage expenses with structured categorization
- Maintain ingredient-level inventory accuracy
- Generate management reports instantly
- Deliver AI-based forecasting and advisory insights

2. Architecture and Stack

Flow: React Dashboard -> Express API -> Business Services -> MongoDB -> AI Processing Layer

- **Frontend:** React/Next.js, Tailwind CSS, Recharts
- **Backend:** Node.js + Express.js
- **Database:** MongoDB
- **AI:** OCR + LLM + forecasting models

3. Role-Based Access (RBAC)

- **Admin:** tenant management, user control, system configuration
- **Manager:** inventory, procurement, approvals, analytics review
- **Cashier:** order creation, billing, payment recording

4. Functional Modules

Orders and POS

Table-based ordering, invoice generation, and payment settlement with audit trail.

Expenses

Receipt uploads, vendor tagging, category mapping, and correction workflow.

Inventory

Ingredient stock tracking with low-stock alerts and consumption deduction from dish sales.

Reports

Revenue, cost, profit margin, and item-level performance dashboards for decision-making.

5. AI Integration

Smart Expense Categorization

OCR extracts receipt text, AI classifies line items, and accounting categories are populated automatically.

Sales Forecasting

Historical order patterns are analyzed to predict next-week demand, peak hours, and high-volume dishes.

Business Insight Engine

AI flags margin pressure and cost anomalies, helping teams react quickly.

AI Chat Assistant

Natural language queries such as "What was today's net profit?" return direct answers from business data.

6. API Design Summary

- /api/auth/* for login, registration, and profile
- /api/orders/* for order operations
- /api/expenses/* for expense lifecycle
- /api/inventory/* for stock tracking
- /api/reports/* for financial analytics
- /api/ai/* for categorization, prediction, and chatbot

7. 10-Week Execution Plan

Research -> UI design -> backend setup -> auth -> orders -> inventory/expenses -> frontend integration -> AI integration -> testing -> deployment/documentation.

8. Startup-Ready Enhancements

- Multi-restaurant tenancy
- Cloud deployment (AWS/Vercel/DigitalOcean)
- Subscription billing via Stripe/Razorpay

- Automated weekly email reports
- Security hardening and audit logs

9. Outcome

This project demonstrates full-stack delivery, product architecture, and practical AI application. It is suitable for final-year evaluation and scalable enough to evolve into a startup MVP.

Prepared documentation variant: Branded (Logo + Theme Colors)